

A True Random Number Generator Based on Semiconductor-Vacancies Junction Entropy Source and Square Transform Method

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Abstract—In this work, we demonstrate a high-throughput true random number generator (TRNG), which passes the National Institute of Standards and Technology (NIST) 800-22 and 800-90B randomness tests. A novel TiN/HfO₂/n-Si structured semiconductor-vacancies (S-VOs) junction entropy source is developed with an area of 4 F². The TRNGs are implemented by the proposed square transform method in the voltage domain using an analog-to-digital converter (ADC). According to the measured statistical characteristics, the TRNGs exhibit excellent robustness and a high throughput of 1.6 Mbps by experiment and 20 Gbps by theory.

Index Terms—Entropy source, oxygen vacancies (VOs), true random number generator (TRNG).

I. INTRODUCTION

TRUE random number generator (TRNG) is an essential module in digital circuits and is widely used in cryptography [1], [2], stochastic computing [3], [4], and simulations [5]. Various TRNGs have been demonstrated so far by utilizing various generation methods including CMOS metastability [6], [7], and random processes in emerging devices [8], [9]. Compared with methods using binary random source [10] or current difference between two devices [11], using a counter

to transform random processes, which can be considered as a square transform in the time domain, is a more stable way to generate random numbers [12]. However, the generating time of such a random number sequence is uncontrollable, which limits its application in digital circuits [13]. As a result, a square transform method in the voltage domain, as well as a high random entropy source, is critical for achieving high throughput and robust TRNGs.

In this work, we demonstrate a novel TRNG consisting of a semiconductor-vacancies (S-VOs) junction-based high throughput entropy source and a self-developed analog-to-digital converter (ADC)-based square transformer in the voltage domain. Contrary to other TRNG solutions that might be highly sensitive to environmental conditions or manufacturing variations, leading to compromised randomness in their output, our TRNG solution is insensitive to the degradation of the key devices in the entropy source. In addition, a novel S-VOs-based entropy source is employed, ensuring excellent robustness, controllable generating time, and reliable operation across various environments. Tests of the National Institute of Standards and Technology (NIST) 800-22 and NIST 800-90B are also passed for the demonstrated TRNGs.

II. ENTROPY SOURCE AND THE TRNGS

The S-VOs junction with the TiN/HfO₂/n-Si structure is shown in Fig. 1(a). Three different splits of devices fabricated on three samples are used to study the impact of substrate doping and oxide materials on the performance of TRNG. The substrate doping concentration of Sample 1 (S₁) and Sample 2 (S₂) is 10¹⁶ cm⁻³, while that for Sample 3 (S₃) is 10¹⁵ cm⁻³. To fabricate these devices, 10 nm thick HfO₂ on (S₁ and S₃) and Al-doped HfO₂ on S₂ are formed by atomic layer deposition on the n-Si substrates, followed by 50-nm-thick TiN deposition by sputtering as top electrode (TE). Fig. 1(b) shows the cross-sectional transmission electron microscope (TEM) image of the device. The interface layer consists of an ultrathin native SiO₂ oxide. During the forming process, this layer experienced breakdown, resulting in a negligible impact on electron transport. The diode-like

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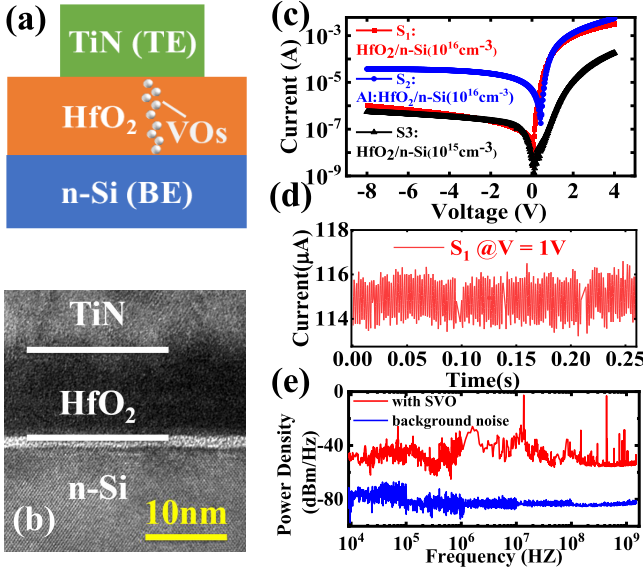


Fig. 1. (a) Schematic structure. (b) Cross section TEM image of the S-VOs junction. (c) I - V characteristics of junctions from three different samples. (d) Current noise of the device from S_1 (1 kHz frequency sampling). (e) Power density of SVO noise and background noise from the spectrum analyzer.

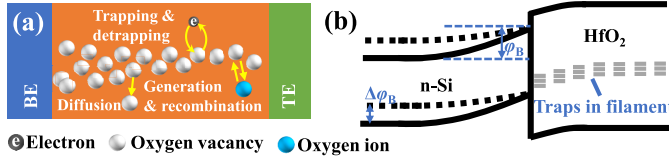


Fig. 2. (a) Origin of randomness in the S-VOs junction. (b) Band diagram of S-VOs junction with trapping effect.

I - V characteristics for devices on different samples are shown in Fig. 1(c). As shown in Fig. 1(d), the measured junction current fluctuates under a constant voltage bias with 1 kHz frequency sampling, indicating that the S-VOs junctions can be used as the entropy sources. Fig. 1(e) shows the power density of SVO noise and background noise from the spectrum analyzer, which indicates that the SVO device can operate in the high frequency condition.

A. Mechanisms of Random Noise in S-VOs Junction

The junction current is dominated by the electron transport between filaments of oxygen vacancies (VOs) and Si interfaces [14]. As illustrated in Fig. 2(a), similar to resistive random access memory (RRAM) [15], the randomness of the S-VOs junction is originated from the physical processes including generation, recombination, and diffusion of the VOs, and the electron trapping/detrapping in these VOs. In addition, these processes will also affect the band bending at the Si/Oxide interfaces [Fig. 2(b)], which can aggravate the fluctuation of the junction current hence enhancing the randomness of the S-VOs junction.

B. ADC-Based TRNGs and Square Transform Method

As shown in Fig. 3(a) and (b), a TRNG prototype has been developed to verify the feasibility of the system. Through

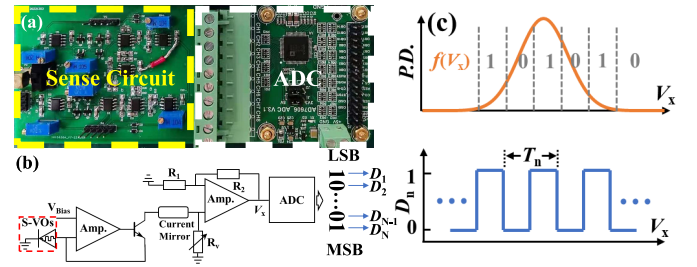


Fig. 3. (a) Photograph and (b) schematic of the developed TRNG. (c) Schematic of the distribution density function of the sense voltage (V_x).

a sense circuit with active load and amplifier, the current fluctuation of S-VOs junction is converted into the fluctuation of sense voltage (V_x), which is input to the ADC and further converted into a digital signal.

Traditionally, a threshold voltage or current is set to determine whether a random signal can be “1” ($>$ threshold) or “0” ($<$ threshold). The random number stream can be achieved by continuously sampling at the output of the entropy source. As a result, it is sensitive to the voltage variation of the entropy source. However, as shown in Fig. 3(c), most of the physical random process obeys Gaussian distribution. To get robust TRNGs, an ADC is utilized to realize the square transform in the voltage domain (1), which divides the V_x into several periods

$$D_n(V_x) = \begin{cases} 0, & mT_n \leq V_x < mT_n + \frac{1}{2}T_n \\ 1, & mT_n + \frac{1}{2}T_n \leq V_x < (m+1)T_n \end{cases} \quad (1)$$

where D_n is the n th bit of ADC, V_R is the minimum resolution of ADC, and T_n is the resolution period of the n th bit, and it satisfies $T_n = 2^n \cdot V_R$. The corresponding generated random number D_n is “0” in the first half period, and “1” in the second half period. The period increases with the n th bit of the ADC, and its minimum value is dependent on the resolution of the ADC. We provide three equations to describe the three criteria [16] that satisfy randomness, namely, statistical randomness (2), unpredictability (3), and nondeterministic (4)

$$P_{D_n=1}(n) = \int_{-\infty}^{\infty} f(x) D_n(V_x) dV_x = 0.5 \quad (2)$$

$$P(B = B_1) = P(B = B_1 | B' = B_2) \\ S_F = \underbrace{D_n^1, \dots, D_n^k}_{B'} \underbrace{D_n^{k+1}, \dots, D_n^{2k}}_B, \dots \quad (3)$$

$$\lim_{k \rightarrow \infty} P(B = B_1 | B'' = B_1) = 0 \quad (4)$$

where S_F is the full random sequence with infinite length, and B_1 and B_2 are bit-groups (BGs) with specific values, respectively. B and B' are the pre and post BGs, respectively, while B'' is another BG. It is noted that each BG does not have shared bits. As the probability density function [$f(V_x)$] is continuous, the statistical randomness can be guaranteed when the period is small enough. In addition, a sequential correlated physical process would affect the unpredictability and nondeterministic of the TRNGs. Using the proposed

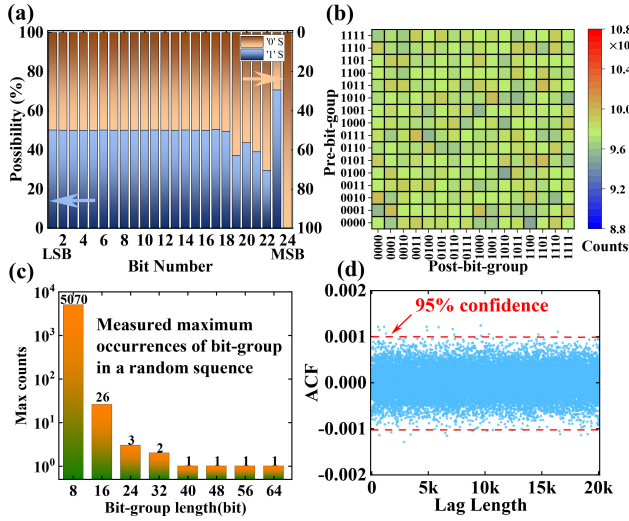


Fig. 4. (a) Measured possibility of “1” and “0” for each bit from 10 M data using 24-bit ADC. (b) Mapping of occurrences of the pre and post BG. (c) Maximum Counts of BGs with different length in the full random sequence. (d) Autocorrelation check results at 95% confidence level.

method makes the “0–1” jump at the least significant bit (LSB) possible even under minor changes of V_x , which minimizes the influence of the sequential correlated physical process.

III. RESULTS AND DISCUSSION

To evaluate the three criteria for the cryptographic security of random numbers, a 10 M-length random sequence from the TRNG is obtained using a 24-bit ADC from S_1 . The possibilities of “0” and “1” for the first 16 bits shown in Fig. 4(a) are almost the same, confirming the statistical randomness of the system (2). Second, the sequence data were divided into 4-bit groups, and the correlation between the neighboring groups was counted and illustrated in Fig. 4(b). The almost uniform correlation mapping exhibits the unpredictability of the generated sequence. Finally, maximum occurrences of bit groups with different lengths are shown in Fig. 4(c). The result shows when the bit group is long enough, it cannot be reproduced (4), which confirms the nondeterministic of the TRNGs. Furthermore, Fig. 4(d) shows an excerpt of the 20 K bit binary stream collected from LSB for autocorrelation with a 95% confidence level. The spatial autocorrelation function (ACF) test shows no correlation among the 20 K bits, which further confirms the randomness.

The current distribution of devices from S_1 to S_3 is measured and normalized as shown in Fig. 5(a). It is found that, under the same bias voltage, the coefficient of variation (σ/μ) is dominated by the Si doping concentration [17]. This is because, in the case of breakdown for the three types of devices, higher doping concentrations lead to lower resistance values, resulting in higher substrate currents. Consequently, the mean current increases faster than the variance of the current. According to the measured Minimum entropy using the 16-bit ADC based TRNGs in Fig. 5(b), only 2 bits passed the NIST 800-90B test without the S-VOs junction entropy sources, and at least 6 bits passed the test when including them. It indicated that the entropy sources are critical for enhancing

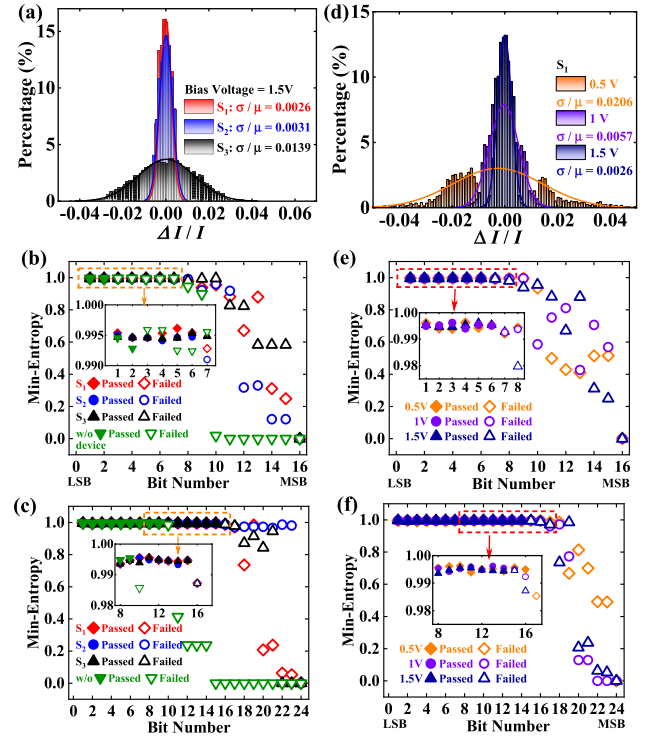


Fig. 5. With different entropy sources of (a) current distribution. (b) Measured min-entropy of 16-bit ADC. $S_1/S_2/S_3$ /without device pass 6/6/7/2 bits. (c) Measured min-entropy of 24-bit ADC. $S_1/S_2/S_3$ /without device pass 14/15/15/9 bits. Under different voltage bias of (d) current distribution and (e) measured min-entropy of 16-bit ADC. 0.5/1.5 V pass 8/6/6 bits. (f) Measured min-entropy of 24-bit ADC. 0.5/1.5 V pass 16/15/14 bits.

the randomness of the TRNGs. In addition, more bits can pass the test with higher σ/μ . As shown in Fig. 5(c), a higher resolution of 24-bit ADC induced more passed bits, proving that smaller T_n in (1) can produce higher quality random numbers. Fig. 5(d) shows the normalized current distributions of the S-VOs junction under different voltage biases. The σ/μ reduces with higher bias voltage. Fig. 5(e) demonstrates that with a larger σ/μ , two more passed bits are observed. Random numbers at different voltage biases were also tested using a 24-bit ADC, and more bits passed the test shown Fig. 5(f). Comparing Fig. 5(e) and (f) further substantiates the more pass bits of higher resolution ADC. In addition, under the condition of significant fluctuations in the mean value, the ability of still obtaining a certain quantity of random numbers demonstrates the robustness of our system. Besides, when random numbers are not needed, the ADC can be used for other analog-to-digital conversion tasks in other hardware systems, maximizing the utility of the ADC.

NIST 800-22 and 800-90B random tests were performed for 100 M bits random sequences from devices on S_1 under a bias of 0.5 V. When using 24-bit and 16-bit ADC-based TRNGs, the 1st–16th and 1st–8th bits can pass all the 800-22 and 800-90B tests, as shown in Tables I and II, respectively. The LSB of the ADC has the smallest V_x period. Therefore, it is easier to pass the statistical random test. Certainly, the ADC with higher resolution can be realized by using more bits, but trade-offs like the corresponding

TABLE I

NIST SP800-22 STATISTICAL RANDOMNESS EVALUATION RESULTS

NIST SP800-2	16bit ADC				24bit ADC	
	1 st bit		8 th bit		16 th bit	
NIST Test	P-Value	Pass Rate	P-Value	Pass Rate	P-Value	Pass Rate
Frequency	0.213	0.99	0.883	1.00	0.740	0.99
Block Frequency	0.534	0.99	0.122	0.98	0.991	0.97
Cumulative Sums-1	0.192	1.00	0.401	1.00	0.367	0.99
Cumulative Sums-2	0.834	0.99	0.924	0.99	0.834	1.00
Runs	0.554	1.00	0.699	0.98	0.145	0.97
Longest Runs of Ones	0.741	0.99	0.739	1.00	0.163	0.99
FFT	0.679	1.00	0.616	1.00	0.137	0.97
Rank	0.048	1.00	0.237	1.00	0.739	1.00
Universal Statistical	0.946	0.99	0.086	0.98	0.719	0.99
Approximate Entropy	0.291	0.98	0.076	0.99	0.115	0.97
Non Overlap. Temp.	PASS	PASS	PASS	PASS	PASS	PASS
Overlapping Temp.	0.739	0.97	0.455	1.00	0.759	0.98
Random Excursion	PASS	PASS	PASS	PASS	PASS	PASS
Random Exc. Var	PASS	PASS	PASS	PASS	PASS	PASS
Linear Complexity	0.816	0.99	0.595	0.98	0.319	0.99

TABLE II

NIST 800-90B TEST RESULTS

NIST 800-90B	Results	NIST 800-90B	Results
Permutation	PASS	LRS Test	PASS (Pr-0.959)
Chi-square Independence	PASS (score=1964, dof=2046)	Chi-square Goodness-of-fit	PASS (score=10.73, dof=9)
Restart Test	PASS	Min Entropy	0.9956

throughput, power consumption, and latency should also be considered. The power consumption of the 16-bit ADC is 100 mW, with a latency of 47 ns, whereas the 24-bit ADC consumes 38 mW and has a latency of 76.8 ns. The throughput of the TRNG can be calculated by the passed bits times the ADC speed. When using the 16-bit ADC (200 K SPS) based TRNG, 1.6 Mbps throughput was obtained experimentally. In terms of noise frequency, two different ADCs with a range of 5 V and one sign bit, the random numbers generated by a 24-bit ADC (30 K SPS) and a 16-bit ADC were passed with 16 bits and 8 bits, respectively. This indicates that the 24-bit ADC is sensitive to a variation of $5 \cdot 2^{15}/2^{23} = 0.02$ V, while the 16-bit ADC is also sensitive to a variation of $5 \cdot 2^7/2^{15} = 0.02$ V. Different sampling rates of the ADC show the same sensitivity to a variation of 0.02 V, which suggests that variation in this frequency range is independent of the noise frequency to some extent. In addition, this noise is believed to be related to the thermal noise of SVO, and the frequency spectrum of electronic thermal noise can be considered as white noise within 100 THz [18]. When using the state-of-the-art commercial 12-bit ADC [19], the theoretical throughput can reach as high as 20 Gbps.

IV. CONCLUSION

A novel TRNG with a high throughput entropy source and an improved transform method was realized in this work.

TABLE III

COMPARISON WITH PRIOR TRNG DESIGNS

	This work	JSSC' 2022 [6]	JSSC' 2019 [7]	IEDM' 2021 [8]	VLSI' 2020 [12]
Entropy Source	S-VOs Junc. RTN	CMOS Metastability	RRAM RTN	STNO RTN	FinFET Relaxation
NIST 800-22 Test Passed	ALL	ALL	ALL	ALL	ALL
NIST800-90B Test Passed	ALL	ALL	N/A	N/A	ALL
Cell Structure	1D (4F ²)	4T	4T2R	1T	1T
Throughput (bps)	1.6 M (Exp.) 20 G (Cal.)	2.39 M	N/A	N/A	N/A
Post Processing	No	Yes	No	No	No

The TiN/HfO₂/n-Si structured S-VOs junction was used as the entropy source for its high throughput and true randomness. The square transform method was also developed and implemented with a 24/16-bit ADC. As summarized in Table III, in this work, the 4 F² entropy source is a CMOS-compatible process without an extra mask and the TRNGs passed all the NIST 800-22 and 800-90B tests. Therefore, the TRNG demonstrated is a promising way to realize a high-speed, robust, and CMOS-compatible random number generator.

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